Tensor computations

Comon's exact-rank conjecture for Hankel tensors

Difficulty: challenging **Importance:** interesting to specialist

Status: Lean verified

Rating rationale: Challenging because exceptional Hankel tensors require a uniform comparison of structured and unrestricted exact decompositions; specialist importance reflects this particular structured tensor class.

Lean proof and verification evidence — 18 October 2026

The complete Lean proof establishes the displayed rank equality for every complex Hankel tensor, including zero and exceptional data and unrestricted ordinary factors. The actual Linux run passed Comparator, the default kernel, LeanCert kernel mode and rejection/isolation controls. Independent replay passed all 41 checks and all 139 original package inputs, including 82 Lean sources. Only the three standard axioms occur.

The reviewed exact statement, proof and retained evidence preserve all original quantifiers and source credit. The stronger rank formula below is not separately exported. The resolution and dated notes below remain verbatim historical records; their description of the earlier informal audit predates this formal proof. Exact originals are retained.

Resolution: affirmative, 12 September 2026

Sidney Holden, Center for Computational Biology, Flatiron Institute, Simons Foundation, proves the complete displayed target in Theorem 1.1, Sections 2–5: ordinary and symmetric rank agree for every complex Hankel tensor of every order $m \geq 3$ and dimension $n \geq 2$, including zero and exceptional tensors. No genericity or Vandermonde restriction is imposed. The stronger formula for nonzero tensors is

$$R(H) = R_{\mathrm{sym}}(H) = \min\{D - r + 2, (m-1)r - (m-2)s\}, \qquad D = m(n-1),$$

where r is the middle Hankel catalecticant rank and s counts distinct projective roots of a smallest-degree binary apolar polynomial. The balanced case is independent of its choice. Proof source · Submission, provenance and verified affiliation.

The complete argument passed a separate independent Codex AI-agent audit. This is informal automated review, not external human peer review or formal verification. No Lean verification was performed. The original target and prior-source credit are retained below; ratings above are historical assessments of that target.

Statement

For every $m \geq 3$, $n \geq 2$, and $h \in \mathbb{C}^{m(n-1)+1}$, form the symmetric tensor

$$H_{i_1 \ldots i_m} = h_{i_1 + \cdots + i_m - m}, \qquad 1 \leq i_j \leq n.$$

Is $R(H) = R_{\mathrm{sym}}(H)$ always true? Here $R(H)$ is the least integer $r \geq 0$ admitting

$$H = \sum_{j=1}^{r} u_{j,1} \otimes \cdots \otimes u_{j,m}, \quad u_{j,k} \in \mathbb{C}^n,$$

whereas $R_{\mathrm{sym}}(H)$ is the least r admitting $H = \sum_{j=1}^{r} c_j v_j^{\otimes m}$, with $c_j \in \mathbb{C}$ and $v_j \in \mathbb{C}^n$. Empty sums represent zero. No genericity or Vandermonde restriction is imposed.

Relevance

This would justify symmetry-preserving decomposition lengths for every Hankel tensor, including exceptional data. Such structure appears in harmonic retrieval and structured tensor factorization.

References

1. J. Nie and K. Ye, *Hankel Tensor Decompositions and Ranks*, SIAM J. Matrix Anal. Appl. 40 (2019). DOI; primary preprint, §6, Conjecture 6.3, p.16.
2. L. Qi, *Hankel Tensors: Associated Hankel Matrices and Vandermonde Decomposition*, 2014. Primary preprint, §1, equation (1), and §4, Theorem 3, for Hankel structure and decomposition.

Status check — 2026-09-10

Rechecked Nie–Ye, Conjecture 6.3, Corollary 4.3 and Theorem 5.4, and searched for later Hankel Comon proofs and counterexamples. Generic even-order and order-three Hankel tensors are proved cases of the displayed universal equality; exceptional tensors and the remaining orders are not settled by those results. The paper already accounts for Shitov's unrestricted counterexample. No later full resolution was located.